



VIT<sup>®</sup>

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)



# VIT STELLAR

— PRESENTS —

OFFICIAL-NEWSLETTER

## ANTARIKSHVANI

EDITION- 01

AUGUST



# ABOUT VIT-STELLAR



VIT-STELLAR is VIT's official gateway to the cosmos. Designed for dreamers, tech enthusiasts, and creative minds alike, we bring the universe down to Earth through overnight stargazing marathons, hands-on rocketry builds, and high-tech rover workshops. But our orbit extends far beyond astrophysics. We are a thriving ecosystem where science meets creativity! Whether you are crafting striking visual designs, writing insightful newsletters, producing dynamic media content, or managing flagship event logistics, there is a place for your passion here. At VIT-STELLAR, curiosity fuels adventure, and every member helps spark the extraordinary. Step into our galaxy and let us explore the unknown together!



# FACULTY CO-ORDINATOR



Dr. Selva Rani B

"Some people guide a club.  
Others become a part of its  
story. For me, STELLAR has  
been both"

Over the years, watching STELLAR grow has been one of the most rewarding parts of my journey. What began as a group of curious students has evolved into a family bound together by passion, dedication, and a shared love for learning. Through every event, every challenge, and every achievement, I have had the privilege of witnessing generations of students grow into confident, capable individuals.

What makes this club special is not just its accomplishments, but the people behind them. The enthusiasm, creativity, resilience, and kindness that each member brings continue to inspire me year after year. Being a part of their journey, offering guidance when needed, and celebrating their successes has always been a privilege I cherish deeply.

As STELLAR enters yet another chapter, I am filled with pride in how far it has come and excitement for everything that lies ahead. The faces may change, but the spirit of this club remains constant. I look forward to many more memories, milestones, and moments of wonder with the STELLAR family in the years to come.



# EXECUTIVE BOARD



## Anushka Arora (Chairperson)

“Three years ago, I joined VIT-STELLAR because I loved astronomy. I never expected to find a second home.”

From my time as a Junior Core Member to Co-Secretary and now Chairperson, my favorite memories remain the quiet moments after meetings, event anticipation, and the team that became family. Every challenge brought a shared lesson, proving that no journey is walked alone. As Chairperson, I look forward to carrying forward our core values and creating space for others to find their place under the stars.

## Arya Diwakar (Secretary)

“If there’s one thing STELLAR taught me, it’s that the best memories are usually made five minutes before a deadline, with a team that somehow refuses to panic.”

Starting as a Junior Core Member and serving as Vice-Chairperson shaped my college life through late-night discussions, event-day chaos, and lasting friendships. The unexpected fixes and boardroom banter taught me the value of trust and teamwork. Stepping into the Secretary role, I am excited to tackle new challenges and build experiences we will look back on with a smile.





# EXECUTIVE BOARD



## Ekansh Garg (Vice-Chairperson)

"Some of the most meaningful things in life aren't found in the loudest moments, but in the people, conversations, and memories that quietly stay with you."

What began as simple excitement for something new quickly turned STELLAR into a defining part of my college experience. Working toward common goals, laughing through stressful moments, and celebrating small victories helped teammates become true friends. Taking on the role of Vice-Chairperson, I carry forward these lessons to create meaningful memories with an incredible team.

## Khushii Anand (Co-Secretary)

"Looking back, it's funny how some of the most important journeys begin without you realizing how much they'll mean to you later."

STELLAR grew from a point of curiosity into a comforting space built on kindness and genuine connection. Through every event and discussion, I learned that true leadership often means simply being there for those who trust you. As Co-Secretary, I hope to foster an inclusive environment where everyone feels welcomed, valued, and encouraged.





# EXECUTIVE BOARD



## Zayn Khan (Events Head)

"I don't think I've figured life out yet, but I've learned that good people, meaningful work, and a few unforgettable memories make the journey a lot better."

To me, STELLAR is a space to grow alongside people who bring energy and laughter to every event. Believing in collaboration over working alone, I strive to build an environment where everyone feels comfortably included. As Events Head, my goal is to craft memorable experiences, strengthen friendships, and build moments we will cherish for years.

## Haneesh Yadav (Finance Head)

"I've never been the loudest person in the room, and honestly, that's never been the goal. I've always preferred letting my work speak for me."

Moving from Junior Core to Finance Head taught me the quiet strength of patience, structure, and showing up consistently behind the scenes. Behind every budget and spreadsheet are people and shared dreams worth supporting. I am proud to play my part in keeping our ambitions grounded while helping the club grow step by step.





# EXECUTIVE BOARD



## Ishayu Basu (Technical Head)

"I've always believed that the best questions are the ones that don't have immediate answers."

Curiosity has always been the heart of STELLAR, where asking "why?" is just as essential as finding the answer. Beyond telescopes and technical concepts, the post-meeting conversations and collective effort behind every event taught me invaluable lessons. As Technical Head, I aim to spark that same wonder, making this year full of meaningful questions and shared discoveries.

## Rasheswari Padhi (Design Head)

"I've always been fascinated by the way a single idea can grow into something people remember."

At STELLAR, creativity meets curiosity, turning abstract ideas into tangible designs that tell a story. Though space initially drew me in, the opportunity to experiment, learn from mistakes, and contribute to something larger made me stay. As Design Head, I look forward to creating visuals that capture the spirit of our community and spark wonder in others.





# EXECUTIVE BOARD



## Deeksha V (Publicity Head)

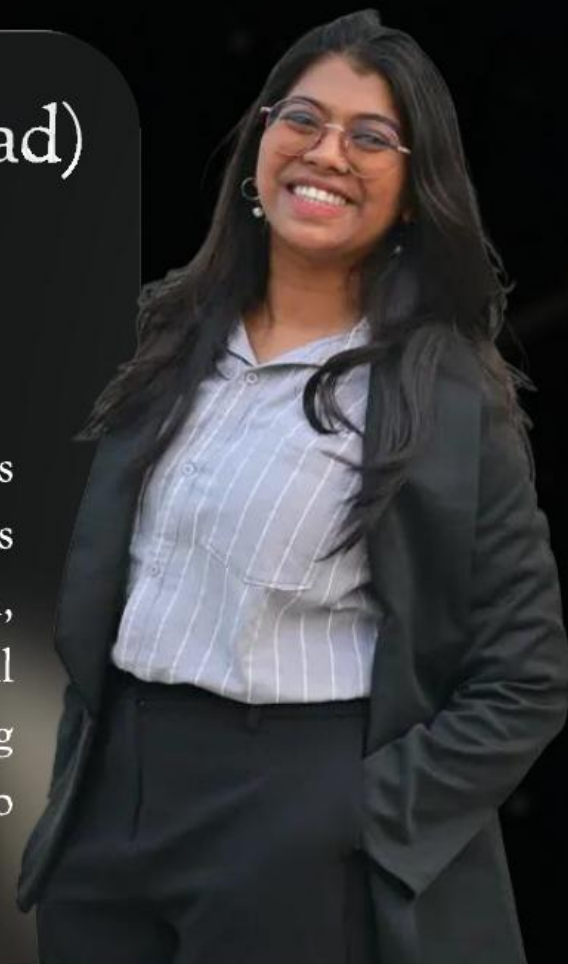
"If there's one thing I've learned, it's that the best experiences aren't measured by how many people see them, but by how many people genuinely feel a part of them."

What started as curiosity grew into a vibrant community where I found confidence, expression, and lifelong friendships. Beyond big events, it is the small conversations and inclusive atmosphere that make STELLAR feel like home. Stepping into the role of Publicity Head, I am excited to share our true spirit and celebrate the people behind our stories.

## Nakshatra Gundeli (Editorial Head)

"I've always believed that the world becomes a little more meaningful when you take the time to understand the stories behind it."

STELLAR has given me a creative space where ideas thrive across writing, art, and quiet conversations. This journey taught me to trust my voice, value quiet growth, and appreciate the people alongside me. As Editorial Head, I aim to nurture a space for storytelling, capturing the moments and perspectives that truly make our club special.





# ADVISORY BOARD



## Subhra Dey (Ex-Chairperson)

**“Somewhere between stargazing, responsibilities, and countless shared moments, VIT-STELLAR quietly became home to me.”**

Starting in 2023 as a Junior Core Member, my journey with VIT-STELLAR shaped me in ways I never expected. Beyond the titles, it was the late-night discussions, shared efforts, and lifelong friendships that made this experience unforgettable. Passing the responsibility of Chairperson to Anushka Arora, I step into an advisory role with complete faith and pride in how far this family has come.

## Sarvesh Baskaran (Ex Secretary)

**“My journey as Secretary has been one of responsibility, growth, and countless moments that made every challenge worthwhile.”**

Serving as Secretary taught me to stay calm through chaos, trust my team, and turn stressful deadlines into fond memories. From late-night event preps to lasting friendships, this club became an integral part of my college life. As I pass the responsibility to Arya Diwakar, I do so with complete confidence, knowing my bond with STELLAR will remain unchanged.





# ADVISORY BOARD

A young man with dark hair and glasses, wearing a white button-down shirt and dark trousers, standing with his hands in his pockets.

## Arya Diwakar (Ex Vice-Chairperson)

“Turns out, being Vice-Chairperson meant spending a year proving that even in complete orbital chaos, things somehow still find their way back into alignment.”

For me, VIT-STELLAR was never just about meetings. It was a space filled with shared laughter, unexpected challenges, and a team that always pulled through. Leadership here meant balancing chaos with calm and becoming someone others could rely on. Handing over the role to Ekansh Garg, I step forward with warmth, gratitude, and memories that will last a lifetime.

A young woman with long dark hair, wearing a light pink button-down shirt and dark trousers, standing with her arms crossed and smiling.

## Anushka Arora (Ex Co-Secretary)

“If there’s one thing being Co-Secretary taught me, it’s that sometimes the real skill isn’t organization, it’s looking completely calm while the universe decides to test your patience.”

What started as a simple role quickly became a deeply personal journey filled with last-minute fixes, shared laughter, and people who constantly inspired me. I learned that leadership is rooted in trust, patience, and showing up when things feel uncertain. As I pass this responsibility to Khushii Anand, I carry forward cherished memories and excitement for the club’s bright future.



# ADVISORY BOARD



## Abishek Ranganathan ( Ex Technical Head)

“From debugging code to decoding the cosmos, I learned that the best technical systems are always powered by shared curiosity.”

Being Technical Head was less about explaining concepts and more about sparking wonder and creating spaces where curiosity brought us together. From late-night sessions to overcoming steep learning curves with an amazing board, this journey was truly special. Passing the role to Ishayu Basu, I am confident that the same passion for cosmic discovery will continue to thrive.

## Aaditya Sharma ( Ex Events Head)

“Event planning taught me that while logistics set the stage, it is the energy of the crowd that actually lights up the room.”

Coordinating events pushed me out of my comfort zone, teaching me to adapt, solve problems on the fly, and trust the process. Beyond the logistics, it was the small victories, shared effort, and incredible people that made every challenge meaningful. As I hand over the reigns to Zayn Khan, I know that energy and creativity will take our events to new heights.





# ADVISORY BOARD



## Parv Pachouri ( Ex Design Head)

“Beyond pixels and palettes, design in Stellar was about giving a visual voice to the infinite wonders of space.”

My time at STELLAR turned a fascination for space into endless hours of visual storytelling. Between chaotic deadlines and endless edits, I realized design was about capturing the emotion and wonder that draws us to astronomy. Passing the role to Rasheswari Padhi, I have complete faith that her creativity will carry STELLAR's story forward brilliantly.

## Anirudh J ( Ex Editorial Head)

“Crafting stories for Stellar showed me that when you give words to the universe, you make the vastness of space feel deeply human.”

Editorial was a space where ideas had room to breathe, helping make the vastness of space feel personal through storytelling. Astronomy is about perspective, and this role allowed me to share that wonder through words, drafts, and quiet conversations. Handing the pen over to Nakshatra Gundeli, I look forward to seeing new stories continue to inspire our community.





# ADVISORY BOARD



## Shivankar Sinha ( Ex Publicity Head)

**“Publicity was never about mere promotion; it was about building a beacon strong enough to gather a community of dreamers.”**

Publicity at STELLAR was about building excitement and creating moments that people truly felt part of. From brainstorming sessions to celebrating successful events, seeing curiosity spark across campus made every late night worth it. As I step back and pass the role to Deeksha V, I know the spirit and reach of this club will only continue to grow.

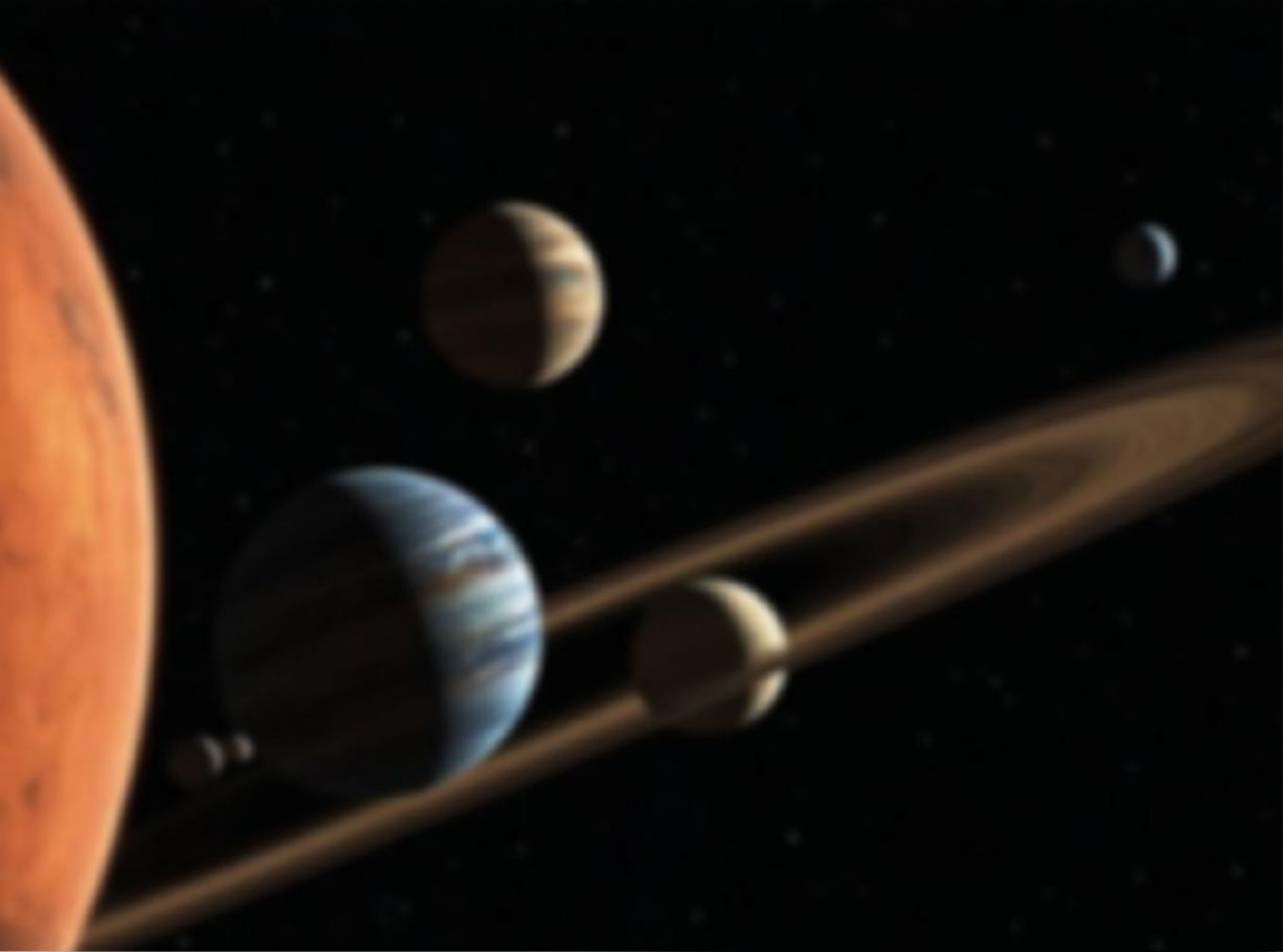
## Rishit Gupta ( Ex Finance Head)

**“Behind every seamless event and bold initiative was a spreadsheet ensuring our highest ambitions had solid ground to take flight.”**

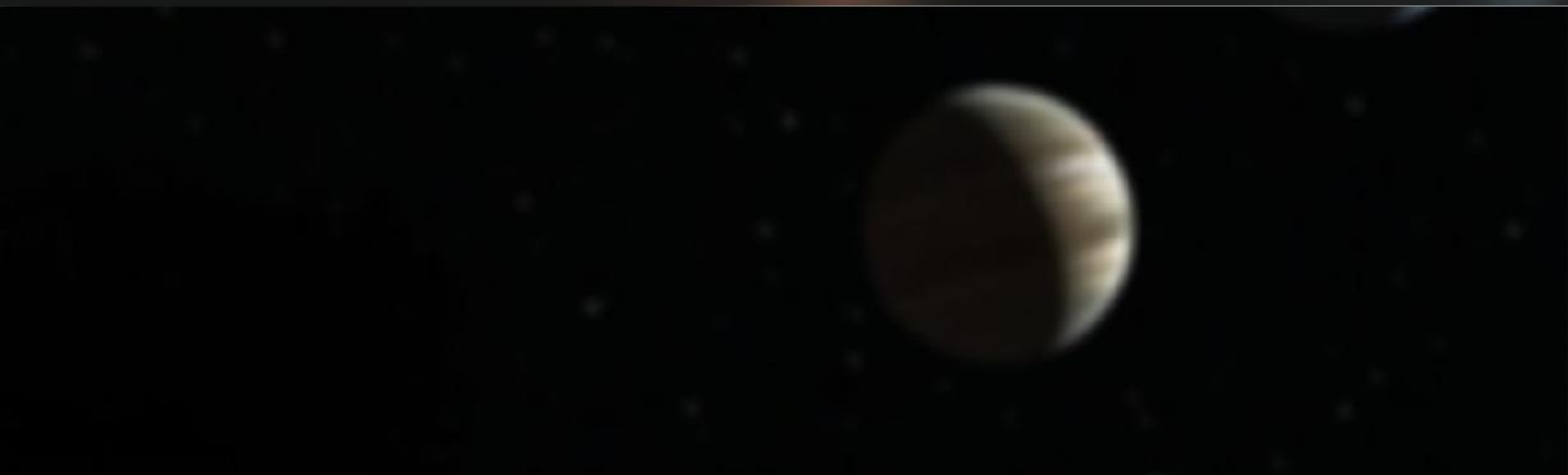
Managing finances taught me how to balance ambitious ideas with practical budgets, but the true value was in the friendships and trust built along the way. Somewhere between nightslips, spreadsheets, and shared laughter, STELLAR became home. Handing over responsibilities to Haneesh Yadav, I step forward knowing the club is in capable hands.







# A LOOK BACK AT THE COSMOS





# CELESTIAL DIVE 4.0

Some experiences shine far too brightly to be forgotten, and Celestial Dive 4.0 was an overnight expedition that truly captured the imagination. Hosted as a flagship stargazing event for graVITas'25 by VIT-STELLAR in collaboration with SpaceTrek India, the event brought together students and astronomy lovers for an all-night exploration running from 9:00 PM to 5:00 AM on September 26th, 2025.

CELESTIAL DIVE 4.0



The event transformed campus grounds into an interactive space observatory where participants engaged directly with observational science. Attendees gained practical experience in optical mechanics by assembling and calibrating small telescopes, practiced specialized astrophotography techniques to capture celestial visuals on their smartphones, and stepped inside an immersive mobile planetarium dome where ancient constellations, zodiac signs, and real-world astronomical concepts came alive overhead.



Although unpredictable late-night clouds eventually forced a pause on deep-sky telescope observations, the energy and enthusiasm across the grounds never faded. The interactive indoor sessions, digital sky-mapping, and shared excitement kept everyone engaged until sunrise, proving that genuine curiosity isn't dependent on clear skies alone.



A sincere thank you goes to SpaceTrek India for their collaboration, to event coordinator Rishmitha Kishore Kumar for her dedication in steering the overnight event, and to all the participants whose passion made the experience unforgettable. Even when clouds hide the stars, our shared curiosity keeps the night sky alive.





# AEROVATE 2.0



Aerospace engineering stepped out of the textbooks and straight into the atmosphere at Aerovate 2.0, VIT-STELLAR's high-octane rocketry workshop hosted on September 23rd and 24th for graVITas. Split across two action-packed days and four intense sessions, the event took participants on an explosive journey from core physics to live launchpad ignition.

The momentum built early at Sarojini Naidu Gallery in SJT. The opening session unpacked the raw fundamentals of aerospace dynamics, giving attendees a masterclass in force, stability, and aerodynamic design. Things leveled up in the second half as participants jumped into software simulation, using OpenRocket to engineer custom fins and bodies, tweaking every micro-variable to optimize flight trajectories.

Day two shifted into high gear at Era Sezhiyan Auditorium in MGB, where virtual blueprints turned into physical reality. Teams of future engineers assembled, glued, and calibrated their custom rockets from scratch. The ultimate payoff came during the final session at the Men's Hostel Outdoor Stadium, where the crowd gathered to watch their creations take to the pad, defy gravity, and streak into the open sky.



This unforgettable event was made possible through the expert guidance of knowledge partners Thrust Tech India Pvt. Ltd. and the dedicated leadership of Event Coordinator Arya Diwakar. Sincere appreciation goes out to sponsors BEBO International, The TJX Companies, Inc., SPAN Inspection Systems Pvt. Ltd., and Farm Guru Organics for backing the vision, and to every participant who brought the high-flying energy that made Aerovate 2.0 a roaring success.





# COSMIC WALK 4.0

Under the quiet canopy of the night sky, Cosmic Walk 4.0 brought stargazers together for an unforgettable journey through astronomy, photography, and community.

The night began at Ambedkar Auditorium, where stargazers gathered for an introductory briefing before embarking on a guided celestial walk across campus. Transforming KC Lawn into an open-air observatory, the technical team worked alongside board members to calibrate high-powered telescopes and guide attendees through the cosmos. Upward-pointed lenses brought the breathtaking details of Jupiter and the cratered surface of the Moon into clear view over multiple observation rounds, offering everyone an intimate look at our celestial neighbors.

While waiting for their turn at the eyepieces, participants soaked in live musical jam sessions that kept the energy vibrant across the lawn. The highlight of the night shifted to interactive sky-mapping as participants learned to navigate constellations using the Stellarium app, right from KC Lawn under the expert guidance of Selva Ma'am. Supported by the dedicated technical team and board members, attendees gained hands-on insight into identifying star patterns and understanding the mechanics of the night sky.





The experience culminated at MB Block with an interactive Astro Pixel photography session, where participants learned long-exposure techniques to capture the beauty of the stars on their own devices. Concluding at SMV Portico, everyone departed with personal astrophotos, fresh astronomical knowledge, and a profound appreciation for the universe above.





# ASTROPIXEL 3.0 WINNERS

Huge Congratulations to the Achievers 🌟



**ALLuri KRISHna**  
**CHaitanya**  
**25BCE0421**



**Haneesh Yadav**  
**24BCE2609**



**AISHani BOSE**  
**25BCE2326**



# SUMMER EVENTS



When campus quietened down for the summer, the momentum at VIT-STELLAR reached an all-time high. Driven entirely by the passion and initiative of the Junior Core, the team executed a remarkable lineup of events throughout the break. This milestone officially established VIT-STELLAR as the second-highest club across the entire institute for events conducted during the summer term.

The series expanded beyond traditional academic talks by seamlessly blending deep theoretical astrophysics with interactive, high-energy engagement. On the scientific front, attendees explored the cutting edge of space science, unraveling complex topics such as black hole sonifications, wormhole physics, dark matter, string theory, and planetary defense strategies for World Asteroid Day. Sessions also celebrated spaceflight history, tracing ISRO's inspiring journey from transporting equipment on bicycles to executing lunar missions, alongside deep dives into Soviet space engineering and Japan's launch programs.



Interactive programming played a central role in keeping participants actively engaged. Stargazers turned into competitors during high-stakes trivia in Cosmic Conundrums, fast-paced strategy challenges like Starfall and The Ultimate Space Battle, and the lighthearted cosmic sketching contest Picasso in Space. This enthusiasm extended seamlessly to social media through interactive Instagram Live broadcasts including Orbital Outwit, Aphelion Arcade, and Astropixel, demonstrating that space education can be as dynamic as a live game show.

Sustaining an event schedule of this scale required exceptional leadership and operational coordination. Special recognition goes to Zayn, Events Head, and Hanesh, Finance Head, for driving this ambitious vision and maintaining momentum throughout the break. Sincere gratitude also goes to all event coordinators and student volunteers who worked behind the scenes to execute every session and make the season a resounding success.





# SUMMER EVENTS-TIMELINE

## MAY

- 17 May: Symphony of the Universe: Data, Resonance, and Sound (Sonakshi Jakhoria & Ayush)
- 19 May: From Earth to Empire: The Future of Civilization (Archismita & Shreyas S)
- 21 May: Wormholes: Physics or Sci-Fi? (Aahana & Ayush)
- 23 May: The Holographic Principle & the Black Hole Paradox (Ayush & Sonakshi)
- 25 May: Beyond the Blue Planet: Multi-Planetary Species (Nischal & Sushant Singh)
- 27 May: Dark Matter: The Ghost in the Machine (Sushant Singh & Saransh)
- 29 May: The Ultimate Space Battle (Sejal Patra & Samarth Sohane)
- 31 May: Cosmic Mystery Challenge (Devapriya Rajesh & Aditi Sooraj Khandelwal)

## JULY

- 1 July: The Illusion of Matter: The Cosmic Symphony (Anutharini & Charan)
- 3 July: Picasso in Space (Sejal Patra & Sushant)
- 4 July: Astropixel Showcase (Aahana)
- 5 July: Aphelion Arcade (Sonakshi Jakhoria)
- 6 July: Stargazing Soirée (Archismita)
- 7 July: Orbital Outwit (Ayush Bhuyan)

## JUNE

- 1 June: Into the Void: Where Space Missions Turned Deadly (Sivaharini & Rakshita Sree)
- 3 June: The Galactic Nexus: Beyond the Speed of Light (Sonakshi Jakhoria & Sejal Patra)
- 6 June: Buran & Energia: The Soviet Giants of Space (Darsh L & Yatheeshwar)
- 8 June: From Bicycles to the Stars: The Story of ISRO (Navya & Soha)
- 9 June: The Cosmic Detective (Sushant & Joshmika)
- 11 June: Starfall (Diya & Saransh)
- 13 June: The God Particle and Its Discovery: Higgs Boson (Nikhil & Aadhyaa)
- 15 June: H-Series: The Backbone of Japan's Space Program (Darsh L & Shreyas)
- 17 June: Where Is Everybody: The Fermi Paradox (Yashwant & Nischal)
- 19 June: Neutrinos: Ghost Particles of the Universe (Viraj & Aadit)
- 21 June: The Search for the Fifth Force of Nature (Samyuktha BR & Vedanth V)
- 23 June: Parallel World and Silent Skies (Arya & Archismita)
- 25 June: Cosmic Conundrums: Space Trivia Showdown (Krishh, Mohit Gandhi & Sanjana)
- 27 June: Voice of the Cosmos (Rakshita & Yaksh)
- 29 June: Beyond Black Holes: Cosmic Extremes (Jahnvi & Shaurya)
- 30 June: World Asteroid Day Special Session (Sonakshi & Archismita)





# ASTRAL CHRONICLES





# DARK MATTER & DARK ENERGY

## The Invisible Forces Shaping the Universe

The universe is far more mysterious than it appears. Everything humans can see, stars, planets, galaxies, gas, and dust, makes up only about 5% of the cosmos. The remaining 95% consists of dark matter and dark energy, two invisible components that shape the structure, motion, and future of the universe. Although scientists cannot directly observe them, their effects dominate the evolution of the cosmos.

Dark matter is a non-luminous form of matter that neither emits nor absorbs electromagnetic radiation. Because it does not interact with light, telescopes cannot detect it directly. However, its gravitational influence on visible matter provides compelling evidence for its existence. One of the strongest examples comes from galaxy rotation curves. According to Newtonian mechanics, stars farther from a galaxy's center should orbit more slowly than those near the core. Instead, astronomers observe nearly constant orbital velocities even at large distances, implying the presence of an enormous halo of unseen mass surrounding galaxies.

Dark matter and dark energy together account for most of the universe's total mass-energy content, yet neither has been directly detected in laboratory conditions. Their existence is inferred through precise astrophysical and cosmological observations.





## Evidence for Dark Matter

### Galaxy Rotation Curves

Stars at the outer edges of spiral galaxies rotate at unexpectedly high velocities, far faster than visible matter alone can explain. This discrepancy suggests the existence of a massive dark matter halo enveloping each galaxy.

### Gravitational Lensing

According to Einstein's theory of general relativity, massive objects bend spacetime and deflect light passing nearby. Astronomers observe galaxy clusters bending background light far more strongly than visible matter permits, revealing large concentrations of invisible mass.

### Cosmic Microwave Background

Minute fluctuations in the cosmic microwave background radiation, the afterglow of the Big Bang, encode information about the composition of the early universe. Measurements from missions such as Planck strongly support models containing dark matter.



# Theoretical Dark Matter Candidates

Researchers continue investigating several theoretical particle candidates through underground experiments, particle colliders, and astronomical observations.

- **WIMPs (Weakly Interacting Massive Particles),** Hypothetical particles that interact primarily through gravity and the weak nuclear force.

- **Axions**

Extremely light particles proposed to solve problems in quantum chromodynamics.

- **Sterile Neutrinos**

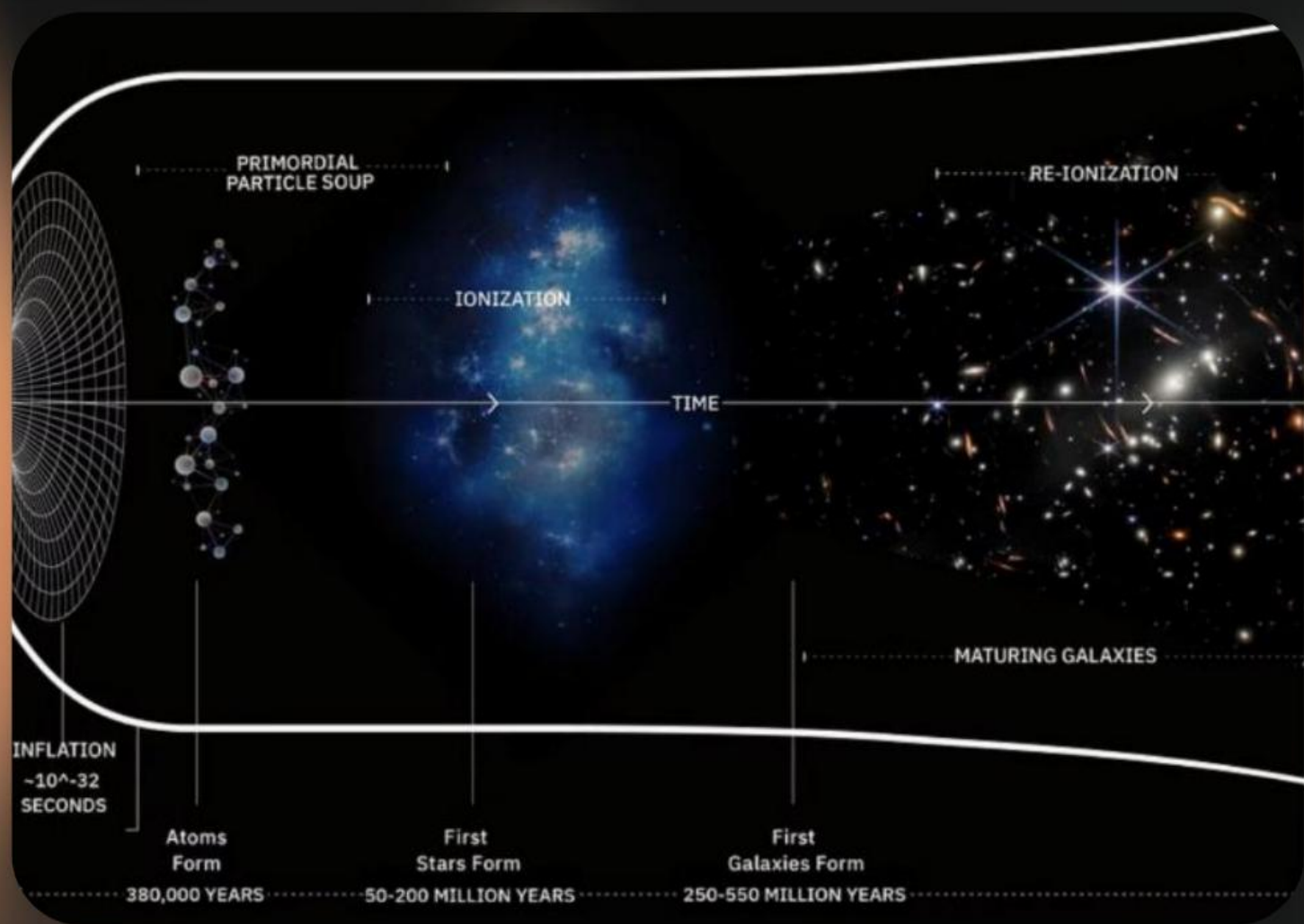
Hypothetical neutrinos that interact only through gravity and possibly through neutrino oscillations.

## The Mystery of Dark Energy

While dark matter acts as cosmic glue holding galaxies together, dark energy appears to drive the universe apart. In 1998, observations of distant Type Ia supernovae revealed that the expansion of the universe is accelerating rather than slowing down. This discovery fundamentally changed cosmology and earned the 2011 Nobel Prize in Physics.

Dark energy is believed to constitute roughly 68% of the universe, while dark matter contributes around 27%. Ordinary matter, everything humans can directly observe, accounts for only about 5%. Despite its dominance, the physical nature of dark energy remains unknown. Some physicists associate it with Einstein's cosmological constant, while others propose dynamic quantum fields or modifications to general relativity itself.





## How Scientists Investigate the Invisible

### Underground Detectors

Facilities such as Gran Sasso Laboratory in Italy and SNOLAB in Canada use ultra-sensitive detectors shielded from cosmic radiation to search for rare interactions between dark matter particles and ordinary matter.

### Space Telescopes

Observatories including ESA's Euclid mission and NASA's WMAP and Planck satellites map the large-scale structure of the universe and analyse the cosmic microwave background to constrain models of dark matter and dark energy.



### **Particle Colliders**

CERN's Large Hadron Collider recreates extremely high-energy conditions similar to those present shortly after the Big Bang, potentially allowing the production of unknown particles associated with dark matter.

### **Galaxy Surveys**

Projects such as the Dark Energy Survey and the Vera Rubin Observatory analyse billions of galaxies to trace cosmic expansion

## **The Road Ahead**

Dark matter and dark energy remain among the greatest unsolved mysteries in modern science. Together they govern galactic structure, cosmic expansion, and the long-term fate of the universe. Future missions and experiments aim to refine our understanding of these invisible components and possibly reveal entirely new physics beyond the Standard Model.

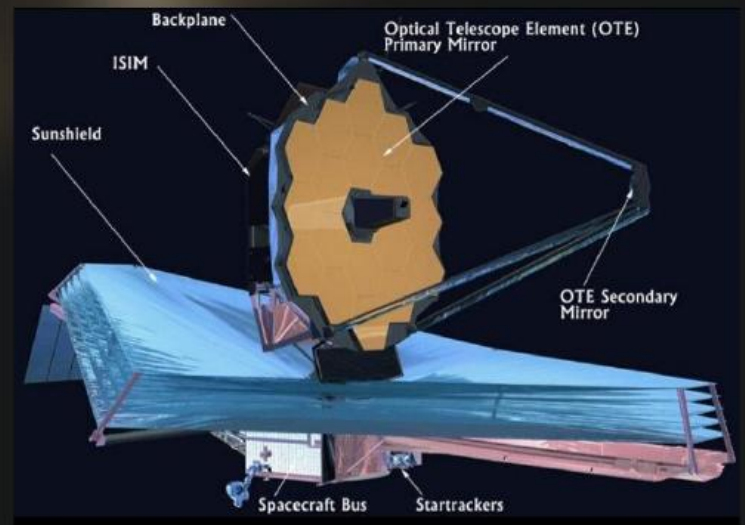
• Ishan Saxena | 25BME0417



# A JOURNEY WITH THE JAMES WEBB SPACE TELESCOPE

The James Webb Space Telescope represents humanity's most ambitious astronomical endeavor. Launched on December 25, 2021, this marvel of engineering orbits the Sun around the second Lagrange point, approximately 1.5 million kilometers from Earth. Its primary mirror, composed of 18 gold-plated beryllium hexagonal segments, spans 6.5 meters in diameter, dwarfing its predecessor Hubble's 2.4-meter mirror.

Unlike Hubble, which primarily observes in visible and ultraviolet light, Webb specializes in infrared astronomy. This capability allows it to peer through cosmic dust clouds, observe the most distant galaxies, and study the atmospheres of exoplanets. The telescope's instruments operate at temperatures near absolute zero, protected by a tennis court-sized sunshield that blocks heat from the Sun, Earth, and Moon.



## Fun Fact

James Webb's sunshield is about the size of a tennis court. It keeps the telescope extremely cold (around  $-233^{\circ}\text{C}$ ), allowing it to detect faint infrared light from very distant stars and galaxies.



# Webb vs Hubble

## Decoding the Red shifted Universe

### 1. The Hubble Space Telescope: Proving the Expansion

The telescope was named after Edwin Hubble, the astronomer who used redshift in the 1920s to discover that the universe is expanding.

The Goal: The Hubble Space Telescope's primary mission was to measure the "Hubble

Constant", the precise mathematical rate at which the universe is expanding.

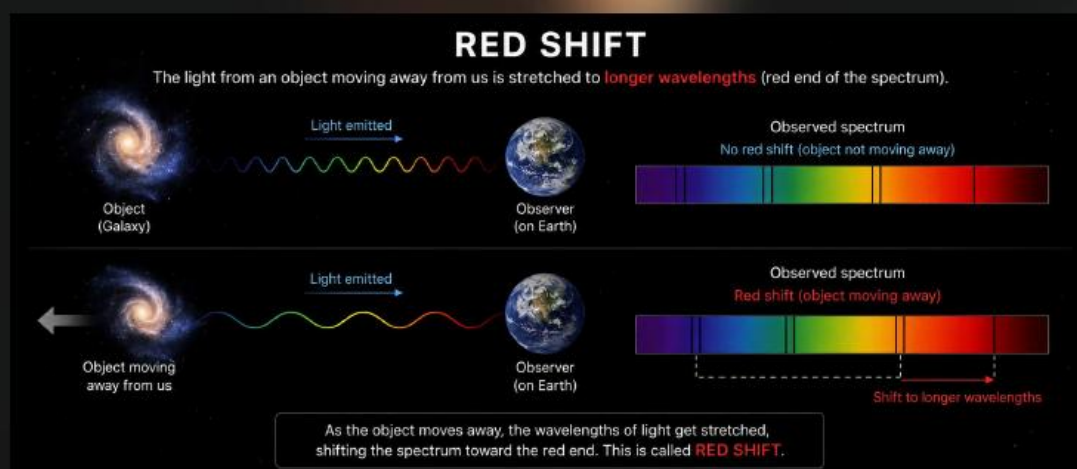
The Limit: As Hubble peered deeper into space, it hit a physical wall. Because the universe is expanding, the farther Hubble looked, the faster those objects were moving away, stretching their light. Eventually, the ultraviolet and visible light from the earliest galaxies stretched so much that it shifted entirely into infrared wavelengths, becoming invisible to Hubble's primary cameras

### 2. The James Webb Space Telescope: Built for Redshift

JWST is the direct scientific successor to Hubble, explicitly engineered to overcome Hubble's redshift limitation.

The Infrared Solution: Because the light from the first stars has been stretching across expanding space for 13.6 billion years, it arrives at Earth as low-energy infrared light. JWST doesn't look at visible light; it uses advanced gold-coated mirrors and cryogenically cooled sensors to look at the universe strictly in the infrared spectrum.

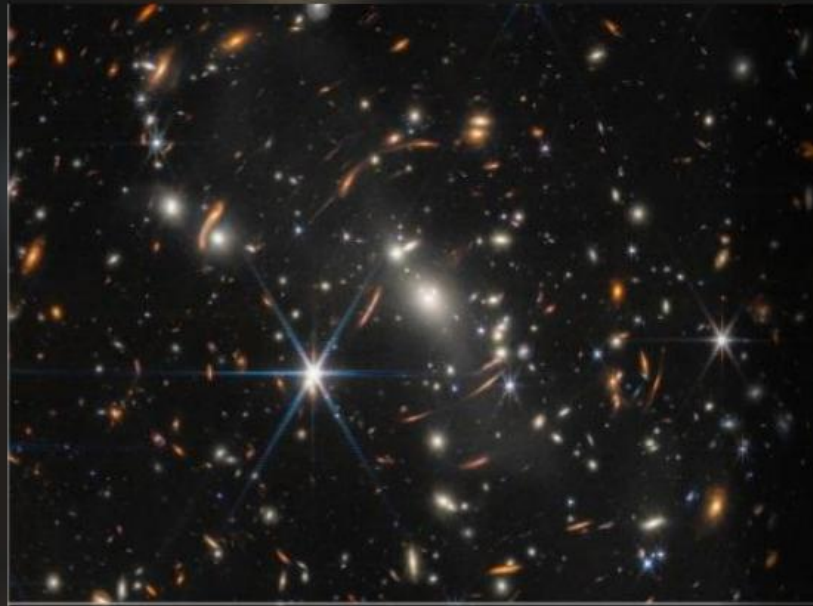
Time Machine Effect: By capturing this highly redshifted infrared light, JWST acts as a cosmic time machine, allowing astronomers to see the universe just a few hundred million years after the Big Bang.



First Deep Field Image



Webb's First Deep Field (the SMACS 0723 galaxy cluster image) was first revealed to the public by U.S. President Joe Biden on July 11, 2022. The official high-resolution data release to the global scientific community followed shortly after on July 12, 2022



Awaiting The Unknown

"SOMEWHERE, SOMETHING INCREDIBLE IS WAITING TO BE KNOWN. THE WEBB TELESCOPE IS OUR WINDOW TO DISCOVERIES BEYOND IMAGINATION."

• Harinipriya K | 25MIS0318

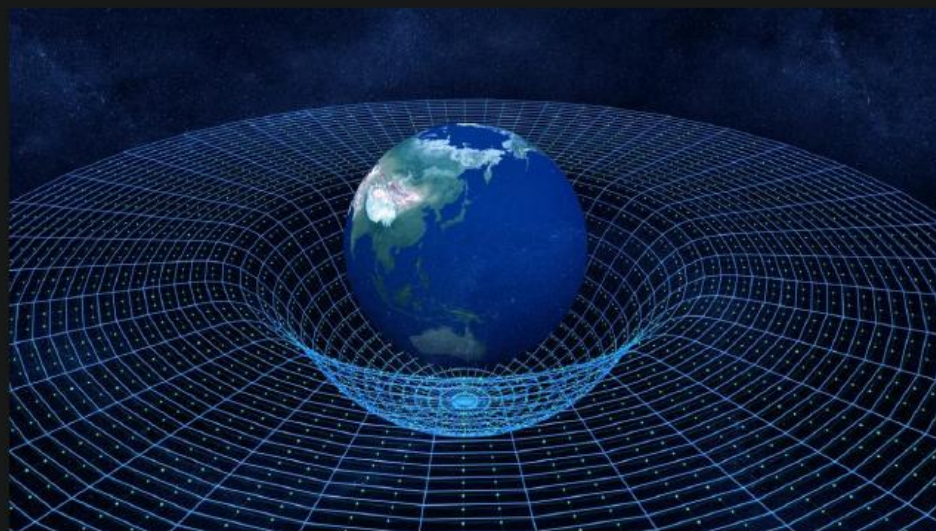


# THE UNIVERSE HAS RIPPLES

## Can space actually ripple?

We usually imagine space as dark and vacuum, but what if space can actually shake and ripple like water. Albert Einstein, hundred years back, predicted something interesting through his Theory of Relativity. He said that space and time are connected together as space-time.

Einstein predicted that massive objects in space can bend space-time around them. And when massive events occur in the universe, they could create ripples in space-time. These ripples are gravitational waves.



## What are gravitational waves?

If a stone is thrown into water, it creates concentric circles isn't it?

In the same way, when an extremely massive object moves in space, they create disturbances all over the space. But it is not like water rippling, it's the whole space time rippling.

Gravitational waves travel at the speed of light and they carry information that a cosmic event that created them.





### **What creates these ripples?**

The most powerful events in the universe can create gravitational waves. A good example of this is when neutron stars crash into each other. A large amount of energy is released, and these events disturb space-time and create ripples that travel across the universe. Supernova explosions can also generate these ripples. Even if these happen billions of light-years away, the waves still reach Earth.

### **Einstein was right**

Gravitational waves remained undiscovered for a very long time because it wasn't easy to detect them. They lacked evidence of existence before 2015, after which scientists found that gravitational waves are actually real and proved Einstein's prediction was correct.

This achievement in modern physics became the greatest one ever, because it showed that space-time can truly ripple.

### **How Gravitational Waves Changed Astrophysics**

Life before this discovery was very difficult because scientists used to observe universe using light through telescopes. When gravitational waves were found, it changed astrophysics forever. Scientists instead of observing cosmic events through light started detecting them through ripples in space-time. Many believe that in future these gravitational waves may help uncover mysteries which are still hidden in the universe.



### **Listening to the universe**

Until today, discovery of this gravitational waves is one of the most fascinating discoveries in modern physics .Even from earth, being billions of light-years from an actual cosmic event can be known to us not by watching, but by listening to these ripples which are incredibly tiny (smaller than the width of a proton) when they reach us. It is amazing that events happening billions of light-years away can still be found on earth by these waves...

• **M Harinivethitha | 25MID0054**



# LIFE BEYOND EARTH

Have you ever looked up at the night sky and felt a sudden, profound sense of smallness? It's a universal human experience. We stare into that vast, glittering ink and inevitably ask the same question: Is anyone else out there?

## What We Know Right Now: The Current Findings

For decades, our search for life was a bit like looking for a needle in a cosmic haystack, using a flashlight with dying batteries. But things have changed. We aren't just guessing anymore; we are gathering hard evidence.

### Mars: A Former Oasis:

Our rovers, like Perseverance and Curiosity, aren't just taking pretty pictures. They are drilling into ancient lake beds. While we haven't found a fossilized Martian yet, we have found complex organic molecules, the building blocks of life. Billions of years ago, Mars had blue skies, rushing rivers, and the perfect conditions for life.

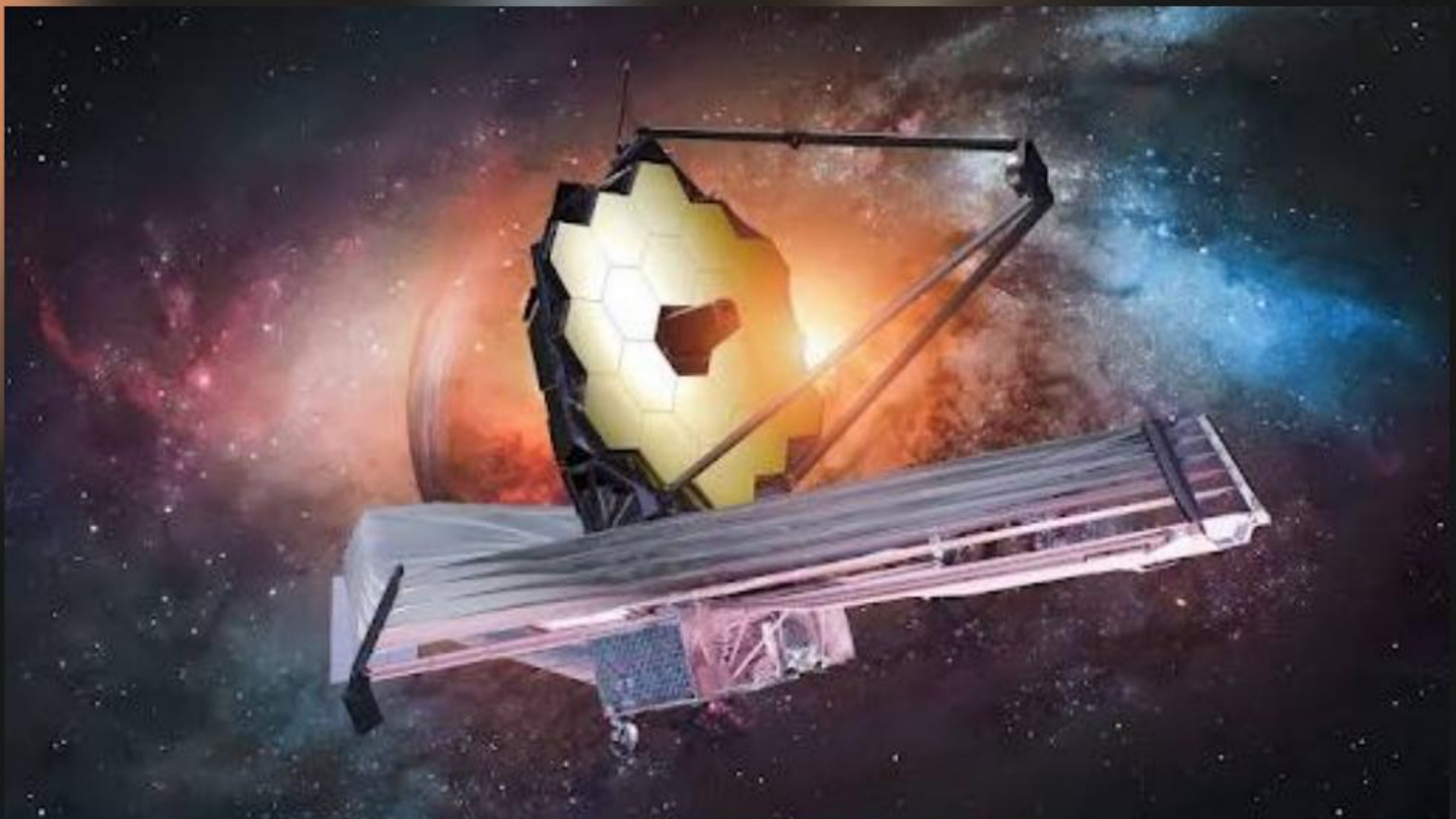


Alone on a red desert, searching for company. Perseverance is digging into Mars' history to find out if Earth is truly the only planet to ever call itself home.



## Webb's Chemical Eyes:

The James Webb Space Telescope (JWST) is currently peering into the atmospheres of distant exoplanets. Recently, it detected hints of carbon-bearing molecules (and controversially, a possible signature of dimethyl sulfide, which on Earth is only produced by living things) in the atmosphere of K2-18b, an exoplanet 120 light-years away. It's not a definitive "yes," but it's a massive "maybe."



Finding our cosmic neighbors. JWST is scanning distant exoplanets, bringing us closer than ever to answering the ultimate question: Are we alone?

## What Could Lie Beneath:

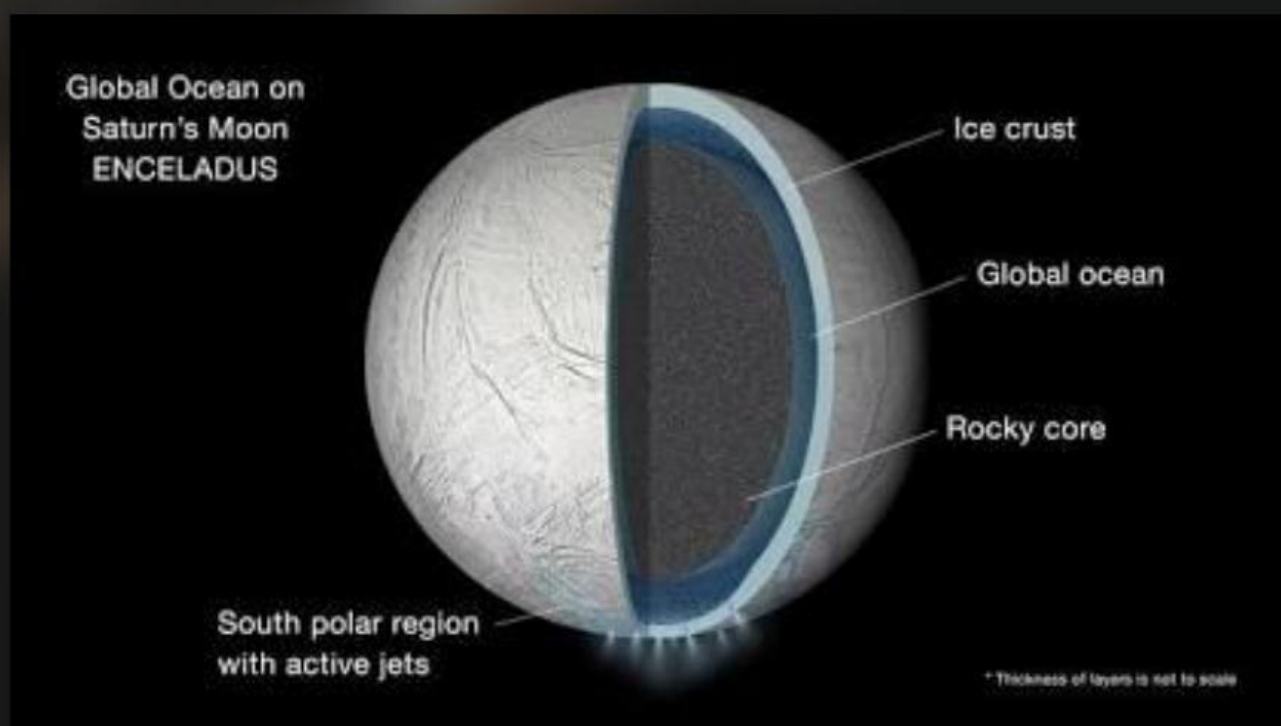
The Ice Moons Here is where the story gets really wild. For a long time, we thought life needed to be on the surface of a rocky planet, basking in the warmth of a star. We were wrong. Meet the icy moons of our solar system: Jupiter's Europa and Saturn's Enceladus. On the outside, they are frozen, dead worlds. But beneath their



miles-thick ice crusts lie global, liquid water oceans. And thanks to the gravitational tug-of-war they play with their massive host planets, their cores are hot.

Where there is water, heat, and rock, there are hydrothermal vents. On Earth, these dark, boiling vents at the bottom of our oceans are teeming with bizarre life, tube worms, blind crabs, and glowing bacteria, that don't need sunlight to survive. They run on chemistry.

Could the same thing be happening in the pitch-black oceans of Europa or Enceladus? The Cassini spacecraft actually flew through plumes of water shooting out of Enceladus's south pole and tasted the spray. It found salts, organic materials, and silica, exactly what you'd expect from a habitable, hydrothermal environment.



Fire and ice in the deep solar system. Enceladus hides a warm, liquid ocean over a rocky core, checking every box needed to support life. Could alien sea life be thriving beneath the ice of Saturn?

## Why It Matters?

Discovering extraterrestrial life, even at the microbial level, will definitively change the course of human history.



It will seal the truth that life is woven into the very fabric of the cosmos. By looking up and digging down, we are not just searching the dark, we are stepping into our inevitable future as a member of a cosmic family.

**Fact corner:** There might already be "life" on the Moon, and it's practically indestructible. In 2019, an Israeli spacecraft carrying thousands of tardigrades (microscopic "water bears") crashed onto the lunar surface. Because these creatures can survive the vacuum of space, freezing temperatures, and intense radiation by going into a death-like hibernation, they are likely still sitting out there right now, waiting for water to wake them up.

• Anirudh Balaji | 25BAI0120

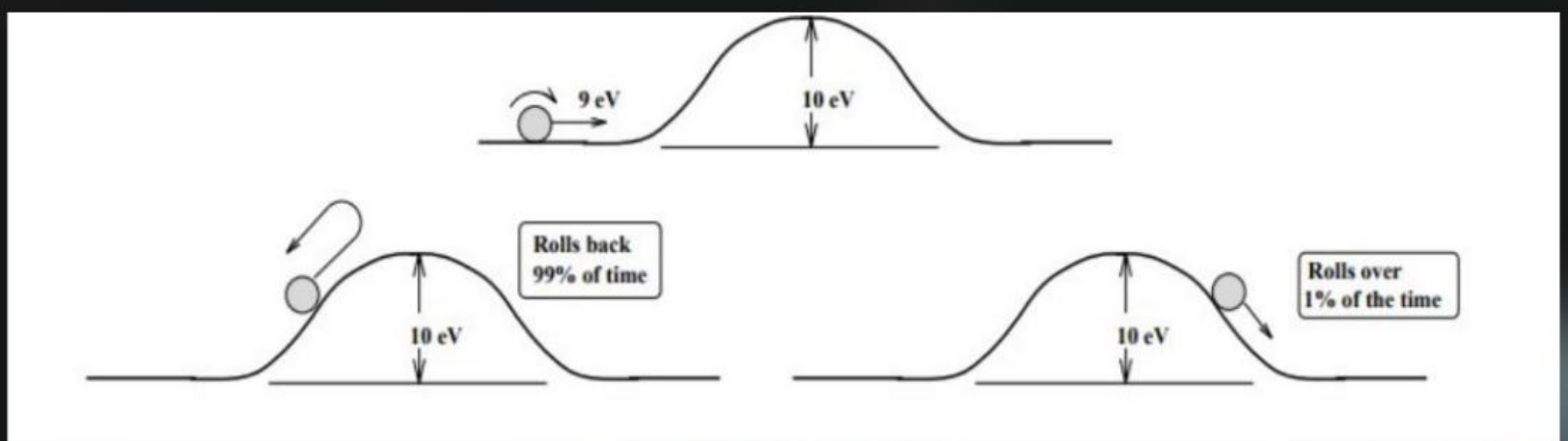


# Quantum Tunnelling

One of the most fascinating powers a superhero can have is teleportation, but theoretically can you actually teleport through matter? Well, we cannot, but small particles can, but only some of the time.

## What is quantum tunnelling?

Quantum tunnelling is a phenomenon that enables low-mass particles, such as an electron, to pass through potential energy barriers, despite seemingly not having enough energy to do so. This is analogous to a ball trying to roll uphill, a ball is a classical particle, and thus it behaves in accordance with classical physics, i.e. it does not have enough energy to overcome the potential energy to roll uphill; but an electron, unlike the ball, is a quantum particle, therefore it can overcome that potential energy barrier, but only some of the time. Therefore, quantum tunnelling seemingly violates classical physics.



Tunnelling is primarily a concept of "Newtonian physics says you don't have enough energy to do something, but Quantum Physics allows it to happen, but only sometimes."



## How does it work?

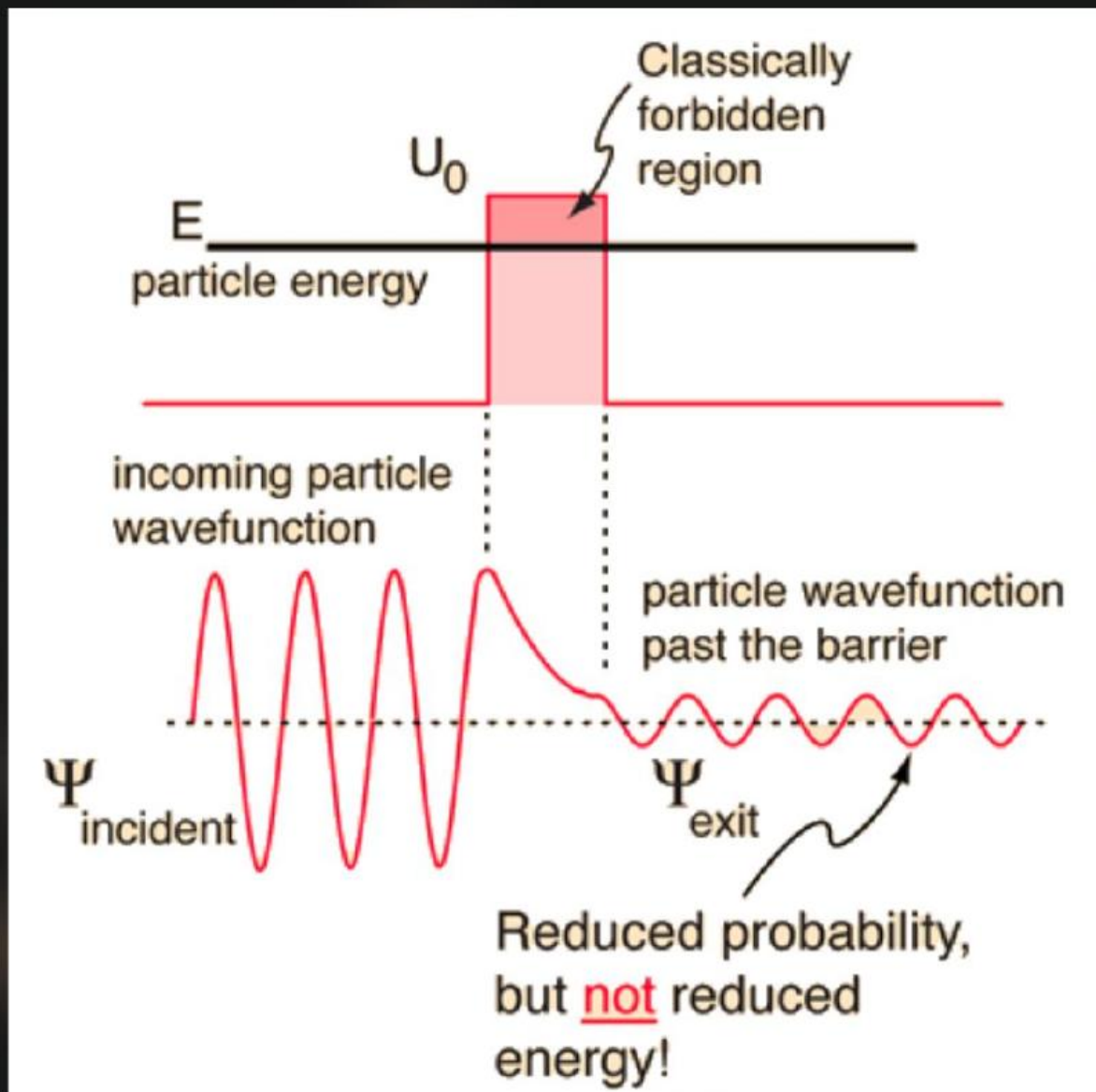
The primary force that stops small, low-mass particles from penetrating through a solid is electromagnetic repulsion; an electron is 'small enough' to fit through the gaps between the atoms in a solid, but the electrons in the said atoms repel the incoming electrons. Quantum tunnelling allows a small portion of the incident electrons to pass through the solid, provided that the solid is a couple atoms thick (the potential barrier is thin enough).

Quantum particles exhibit wave-particle duality, i.e. these particles act like waves instead of solid, localized balls, and are described by wave functions, which allow the calculation of the particles' probability distribution, the probability of where the particle may be at any given moment. When the wave hits a potential barrier, it penetrates in and exponentially decreases within the barrier thickness. Provided, that the barrier is thin enough, a small trail of the probability distribution emerges on the other side of the barrier.

In stars, the main source of fuel is nuclear fusion, smaller nuclei are forced together to form larger nuclei resulting in immense energy being released. In our Sun, positively charged Hydrogen ions,  $H^+$ , collide with each other to form Helium atoms. These  $H^+$  ions experience repulsive forces, thus for nuclear fusion to occur, the ions must be travelling at high enough speeds that they have enough (kinetic) energy to overcome the repulsive forces (potential barrier).

Alas, in reality, the majority of the ions do not travel at high enough speeds, thus according to classical physics alone fusion in the Sun cannot happen at a sustainable rate. Quantum tunnelling enables these ions to gain enough momentum to overcome the barrier, given that the repulsive forces are small enough.





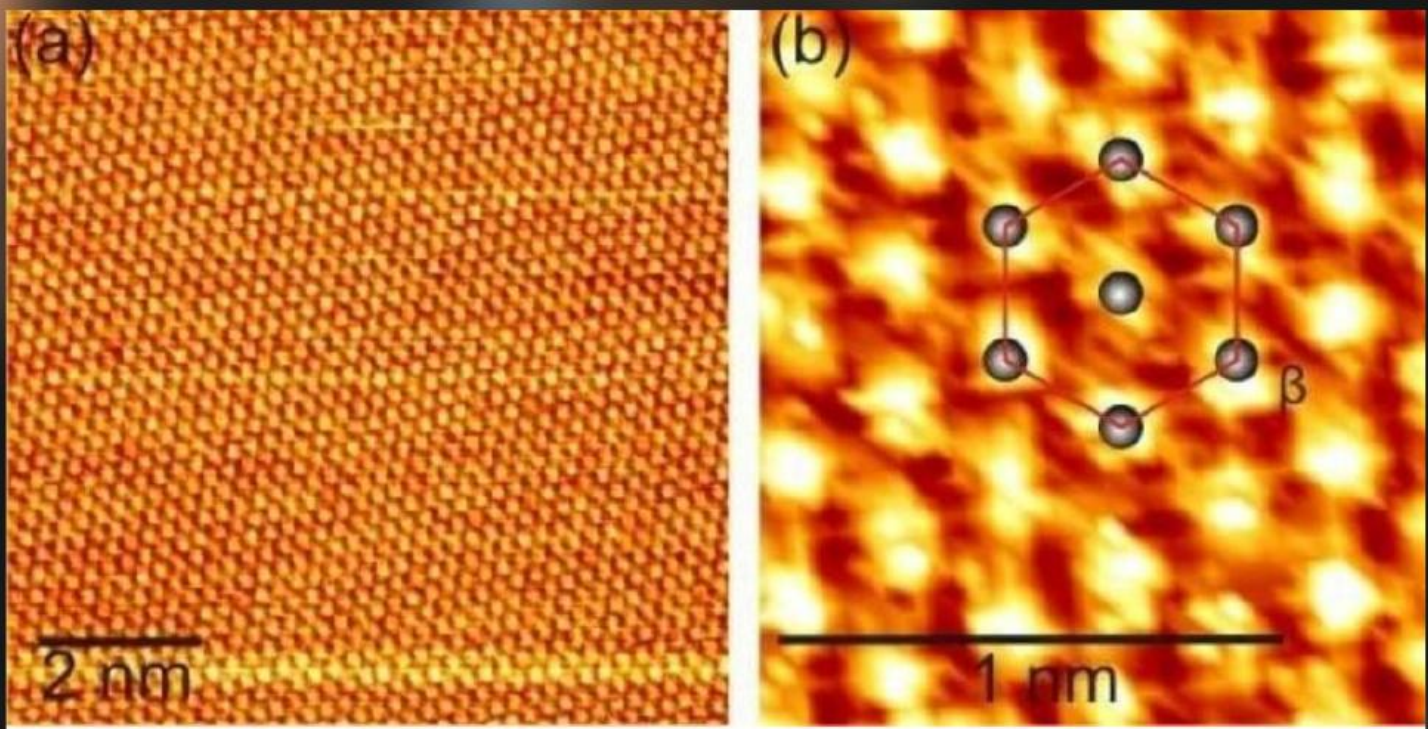
## How is it useful?

Tunnelling plays an essential role in phenomena such as, nuclear fusion, radioactive decay, flash memory, and quantum sensors.

- o Radioactive decay: allows the explanation of decay rates and half-lives; dictates the tendency of alpha particles to tunnel through potential barrier of atomic nuclei to escape.



- o **Flash memory:** enables USB drives, SSDs, and smartphone storage; dictates the tendency of electrons to tunnel through insulating barriers to store data in computer chips. Although, tunnelling limits the size of devices used in microelectronics as electrons readily tunnel through insulating layers and transistors that are thinner than about 1 nm.
- o **Quantum sensors:** used in medical imaging and materials science; tunnelling currents can detect minute changes in magnetic or electric fields.
- o The inherent sensitive nature of the tunnelling process, allows the photographing of individual atoms using scanning tunnelling microscopes (STM).





# THE STRANGE CASE OF TABBYY'S STAR

## THE STAR THAT SHOULDN'T BEHAVE THIS WAY

Somewhere around 1,500 light years from Earth lies Tabby's Star. This star has continued to puzzle scientists for years. Unlike most stars, which dim in predictable ways when planets pass in front of them, Tabby's Star showed us sudden and irregular drops in brightness, blocking around 20% of its light. This unusual behavior caught the attention of astronomers because nothing about it seemed normal. The star appeared to brighten and darken without a clear pattern, making it one of the most mysterious objects ever observed in space.





## ALIEN MEGASTRUCTURE... OR SCIENTIFIC MYSTERY?



As the mystery surrounding Tabby's Star deepened, speculation began to spread far beyond the scientific community. One particularly fascinating theory is the artificial structure built by an advanced extraterrestrial civilization to harness the star's energy. This concept is often referred to as the "Dyson Sphere".

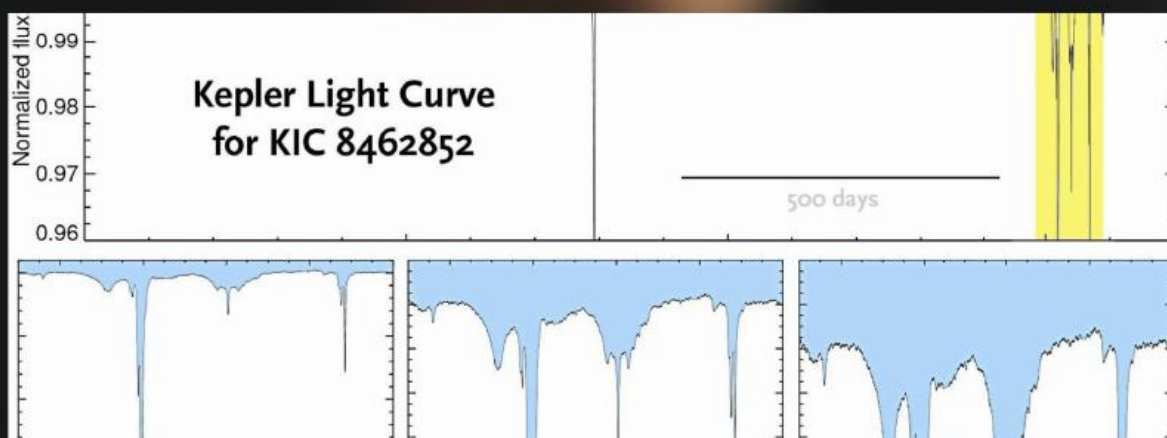
Although scientists never confirmed the possibility of alien life, the star's unpredictable behavior was unusual enough to keep this theory alive. In later studies, researchers proposed more natural explanations, such as dense clouds of cosmic dust or fragmented comet debris. Yet despite years of observation, the true cause behind the star's strange behavior remains uncertain, and I, for one, know that we prefer the less sensible explanations.



# THE ROLE OF AI IN SOLVING COSMIC MYSTERIES

When Tabby's Star first shocked astronomers in 2015, researchers did not have the powerful AI tools that exist today. Scientists had to manually study enormous amounts of telescope data to understand why the star's brightness kept changing in such strange ways. This raises an interesting question: as much as people today fear the rapid growth of AI, can we be completely certain that every possible clue in earlier research was properly identified by humans alone? Now, nearly a decade later, AI systems are being used by NASA and other research organizations to scan millions of space images, discover hidden cosmic anomalies, and identify unusual patterns.

As AI and astronomy continue to merge, companies such as SpaceX, Google DeepMind, and IBM Research are investing heavily in Artificial Intelligence, data analysis, and advanced technologies that could shape the future of space exploration. What makes this even more exciting is that discoveries like these are no longer limited to scientists working in major space agencies, opening opportunities for students interested in this collaboration of AI and astronomy. Platforms like ours, the Stellar club at Vellore Institute of Technology, allow students to discuss real scientific mysteries, explore emerging technologies, and understand how AI could help uncover answers hidden somewhere in the universe. Projects like these create internships, collaborative research, coding projects, and innovation programs where students themselves could contribute to future discoveries.



The irregular dips in brightness observed by NASA's Kepler Telescope made Tabby's Star one of the most mysterious objects in astronomy.



## RESOURCE SPOTLIGHT

NASA JPL, [Mysterious Dimming of Tabby's Star May Be Caused by Dust](#)

NASA Exoplanet Exploration, [A Star with Strange Dimming](#)

NASA, [AI Unlocks Hundreds of Cosmic Anomalies](#)

SETI Institute, [The Mysterious Star KIC 8462852](#)

• Sahila Yazar | 25BCE2642



# UNIVERSE BREAKERS- THE PRIMORDIAL GALAXIES

## Early Universe and Big Bang

The standard cosmological model of the Universe assumes that the properties of Universe is very much same at every point and the laws of physics is universal.

The Universe was born 14 billion years ago from a high-temperature, highly dense state called the hot Big Bang. Our Universe has been expanding and cooling down ever since. All the structures we see around us have resulted from the tiny fluctuations embedded in cosmic inflation, an early period of highly accelerated expansion, which eventually formed galaxies and cosmic clusters under the effects of gravity.

The cosmic microwave background radiation emitted 380000 years after the Big Bang has been travelling the entire universe since then, carrying with it the great historical data that tells the tale of our universe's early origins. As the photons traveled for 13.8 billion years, the fabric of space itself expanded, stretching the wavelengths of light along with it. It shifted the initial warm, yellow-orange visible light into the infrared, and ultimately into the microwave region we observe today through cosmological redshift.

Initially, scientists were only able to observe galaxies that existed 1 billion years after the Big Bang. Going further back in time was difficult, considering technological limitations of earlier telescopes. The light from the earlier galaxies that are located too far away from us has stretched so much that it is no longer in the visible spectrum. James Webb Telescope – a pioneering technology that had capabilities superior to its predecessors, was able to capture the information in this far infrared part of redshifted region. Hence, we were able to look at galaxies formed earlier into universe's timeline that might have been structurally different from the modern galaxies that we see today.



## Red Galaxies in early universe

Galaxies that are actively forming new stars should be bright at short wavelengths because they have lots of giant, hot and short-lived stars emitting a relatively blue color compared to older galaxies.

And older galaxies that are no longer so actively forming stars should look redder because the short-lived blue stars have died off already. And the much older and distant galaxies have undergone cosmological red shift which explains the redness.

Scientists were able to isolate heavy weight red galaxies that have developed 500 to 700 million years of the Big bang which had stellar mass density much higher than anticipated. In essence, according to our current math and models of how the universe began, there shouldn't have been enough time for galaxies to get that big, that fast.

The initial discovery relied on photometry that measures brightness through colours. To confirm that the redness was actually due to the highly evolved stellar populations rather than cosmic dust or active galactic nuclei (supermassive black holes), scientists had to perform spectroscopic follow-up where they break the light into its individual component wavelengths to study the composition of structures in detail.

Once spectroscopy confirmed that some of these enormous and dormant galaxies truly existed so early in time, it threw standard cosmological timelines partially into shambles and chaos. This forced astrophysicists to hunt for extreme and innovative new mechanisms that might have taken place during early galaxy evolution to reconcile these observations with our fundamental understanding of the universe.



## The Search for Explanations

A few major scientific explanations were proposed to resolve the mysterious nature of these galaxies while simultaneously keeping the Big Bang Model intact. Two prominent ones were the OVERLAPPING IMF THEORY and Active Galactic Nucleus Feedback.

## Deep Into Void, Unleashing More Chaos:

If you thought that a universe full of premature, red galaxies is puzzling, look at what happens when we look further back uncovering ancient cosmic structures at the very edge of the observable universe.

### i) The Farthest Galaxy Detected



JWST NIRCam composite image of MoM-z14- Spectroscopically confirmed at a redshift of  $z=14.4$

The galaxy designated MoM-z14 formerly known as JADES-GS-z14-0 is currently the farthest galaxy ever detected, spotted by James Webb Space Telescope's NIRCam (Near-Infrared Camera) and confirmed spectroscopically with its NIRSpect (Near-Infrared Spectrograph) instrument.



We are seeing this galaxy as it appeared in the distant past, only 280 million years after the Big Bang. It is one among a growing group of surprisingly bright galaxies in the early universe with its more compact structure and heavy chemical enrichment. MoM-z14's mature state so early into universe's timeline holds essential clues to understand how different things were back then in the early universe.

## **ii) The Galaxy that stopped Spinning**

XMM-VID1-2075 discovered and confirmed of existing in 2026, is a massive galaxy almost red and dead, formed less than 2 billion years after the Big Bang. It appears to have no rotation at all, a trait usually seen only in much older, evolved galaxies. This poses challenges to current cosmological simulations, which predict that non-rotating galaxies should be extremely rare so early in cosmic history, urging astronomers to rethink the timescales and processes of early galaxy formations.

## **The Next Chapter**

The upside to all of this is the fact that for the first time we have the opportunity to test our theories and run simulations of models alongside real data. Before completely dismissing the Universe's timeline, we should address the gaps in our existing theories of the early universe. The most suitable way of navigating this avalanche of new data is to continue studying the systems behind structure growth, star formation, black hole seeding and growth of galaxies. This will help us re-model our current hypothesis and studies about the early universe. And once we figure things out more vividly, the impossible early galaxies will become a common part of the new and updated understanding of our early spacetime.

They remind us that the more data we gather, the more we must be prepared to let go of old certainties to make room for a complex and accurate reality.



# WHEN BLACKHOLES COLLIDE:

## A Ripple Through Time and Space

We think of a black hole as a void in space, where nothing can escape. In *Interstellar*, the Christopher Nolan movie, we also think that a black hole can be a massive object where time is also a dimension in space where he sends a message to his daughter when she was young. To be honest, no one really knows what happens inside a black hole. Scientists understand that black holes are the result of the collapse of a massive star. However, there seemed to be another case after the detection of an event in 2015. This detection confirmed Einstein's theory of Relativity, changing the way we perceive gravity. What was detected?

### What is a Black Hole Merger?

A black hole is an astronomical object with a gravitational pull so strong that not even light—the fastest moving electromagnetic wave—can escape it. It is a result of the collapse of a massive star after it exhausts its nuclear fuel.

There are some black holes which exist in binary systems. In such systems, two black holes orbit around their centre of mass. Over the course of a billion years, these black holes spiral towards each other and as a result of loss of orbital energy through gravitational radiation, they start moving very fast towards each other, merging into one larger black hole.

This merging creates a ripple in space-time, producing gravitational waves that may take a billion years to reach us.



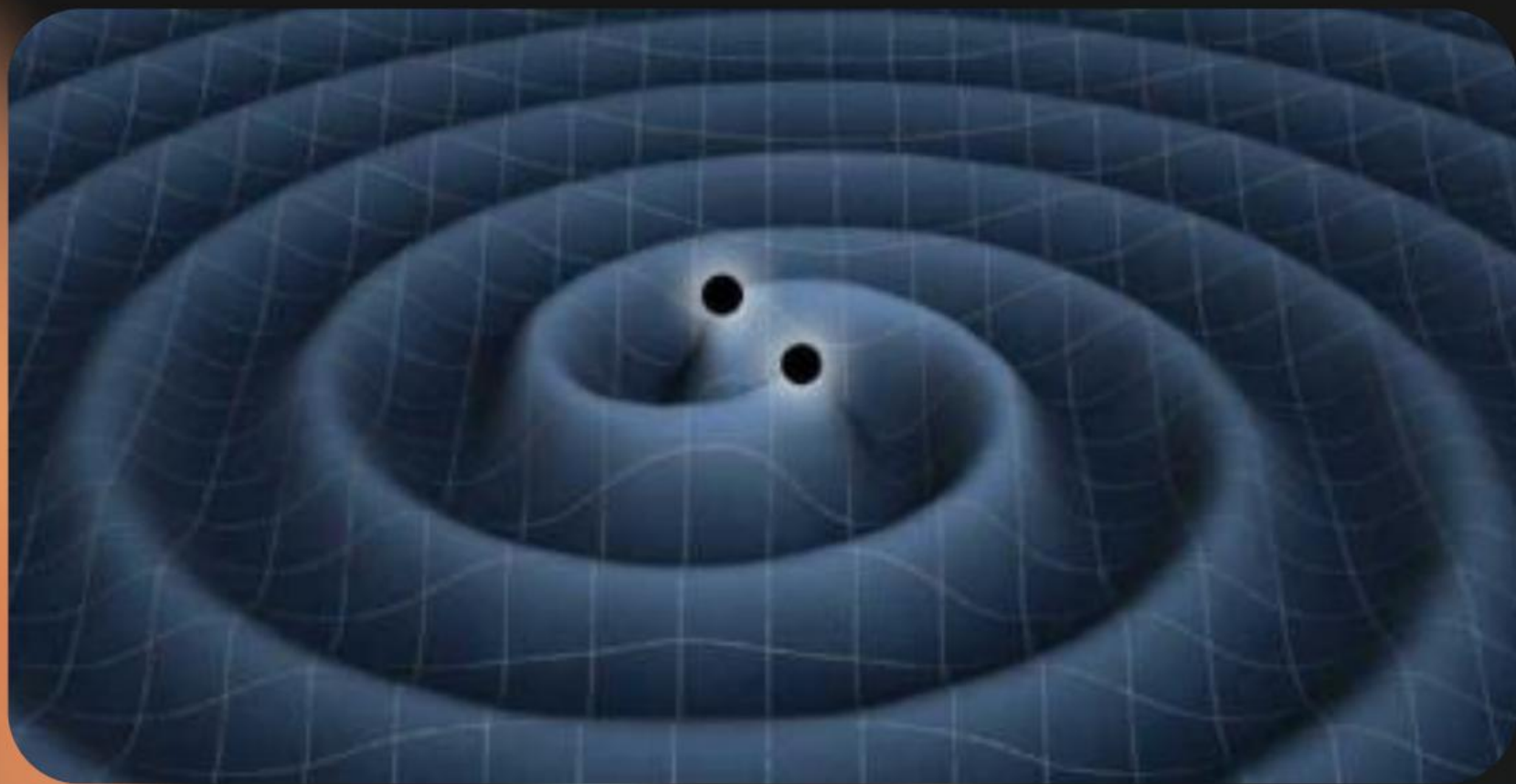


## Gravitational Waves

Einstein's General Theory of Relativity explains gravity as a curvature in space-time rather than a force as explained in classical physics. This can be illustrated by placing a ball in a tablecloth spread horizontally. The tablecloth represents spacetime and the ball represents a massive object. Without the ball, the cloth is normally spread. When the ball is kept, the cloth 'warps' around the ball. Smaller balls can be thrown and they 'orbit' around the bigger ball. These smaller objects represent moons, planets, etc.

Massive objects produce a warping of space and slowing of time. When the black holes start spiralling towards each other, their orbital velocity, i.e., the velocity with which they orbit around each other, increases. As this happens, the orbital energy, i.e., the energy the black hole has to orbit the other one, and the angular momentum start decreasing. This energy is carried away by gravitational waves which move at the speed of light. As the orbital speed increases, the frequency of the gravitational waves increases.





When the black holes finally collide, they emit a burst of gravitational waves. This is considered the most intense and energetic phase of the merger. For example, the event GW150914 produced gravitational waves with the peak power output of  $3.6 \times 10^{49}$  W, which is 50 times the combined power of all light emitted by all the stars in the universe.

After the merging, the merger reaches the phase which is termed the Ringdown phase. After the merger, the larger black hole is extremely unstable and distorted. It goes through a relaxation process where it 'rings' as if someone struck a bell. During this phase, the resultant black hole produces very small gravitational wave vibrations.

While much of the energy is carried away by gravitational waves, mass is also converted into gravitational radiation. During the GW150914 event, approximately 3 solar masses were converted into energy carried by gravitational waves. Simulations of equal mass black hole mergers show that roughly 5% of the total mass and energy is converted into gravitational radiation.



# LIGO

As already discussed, space-time is distorted in such mergers. They send ripples in space-time known as gravitational waves. The flow of time is significantly altered- time flows much slower near the black holes. When they merge, distortion reaches maximum intensity, creating a 'storm' in the very fabric of space and time.

At the Laser Interferometer Gravitational-Wave Observatory, they use highly sensitive instruments to detect the gravitational waves. By the time these gravitational waves reach Earth, they are extremely weak. Such an instrument is known as a Michelson Interferometer.

This measures minute changes in distance between test masses.

When a gravitational wave passes through, it causes space to stretch in one direction and compress in the other, slightly altering the length of the arms of the interferometer. LIGO can detect distortions in space-time that are 10,000 times smaller than the nucleus of an atom, thus making it the most precise ruler to be ever constructed.



## Why these discoveries matter

The discovery of gravitational waves changed the way scientists understood gravity. This discovery confirmed Einstein's theory that gravity can be described as a warping of space-time, leading to a better understanding of the workings of the cosmos.



# Artemis II

## What Artemis II Is

Artemis II was a crewed flyby of the Moon. It was the first crewed flight beyond low Earth orbit since Apollo 17 in 1972, the first crewed flight of the NASA-led Artemis program, the first crewed flight of the Space Launch System (SLS), and the first crewed flight of the Orion spacecraft.

## Mission Goals and Significance

The Artemis II test flight will validate the systems needed to sustain astronauts on a deep space journey and prepare to build a long-term human presence on the Moon. The primary goal of Artemis II is to prove the ability of systems and teams to support the flight crew in the flight environment and their return to Earth.

The Environmental Control and Life Support System (ECLSS) on the Orion spacecraft maintains four astronauts for up to 21 days by controlling cabin conditions, removing contaminants, providing water and gas, and providing essentials such as waste management, thermal control, food warmers and sleeping bags.

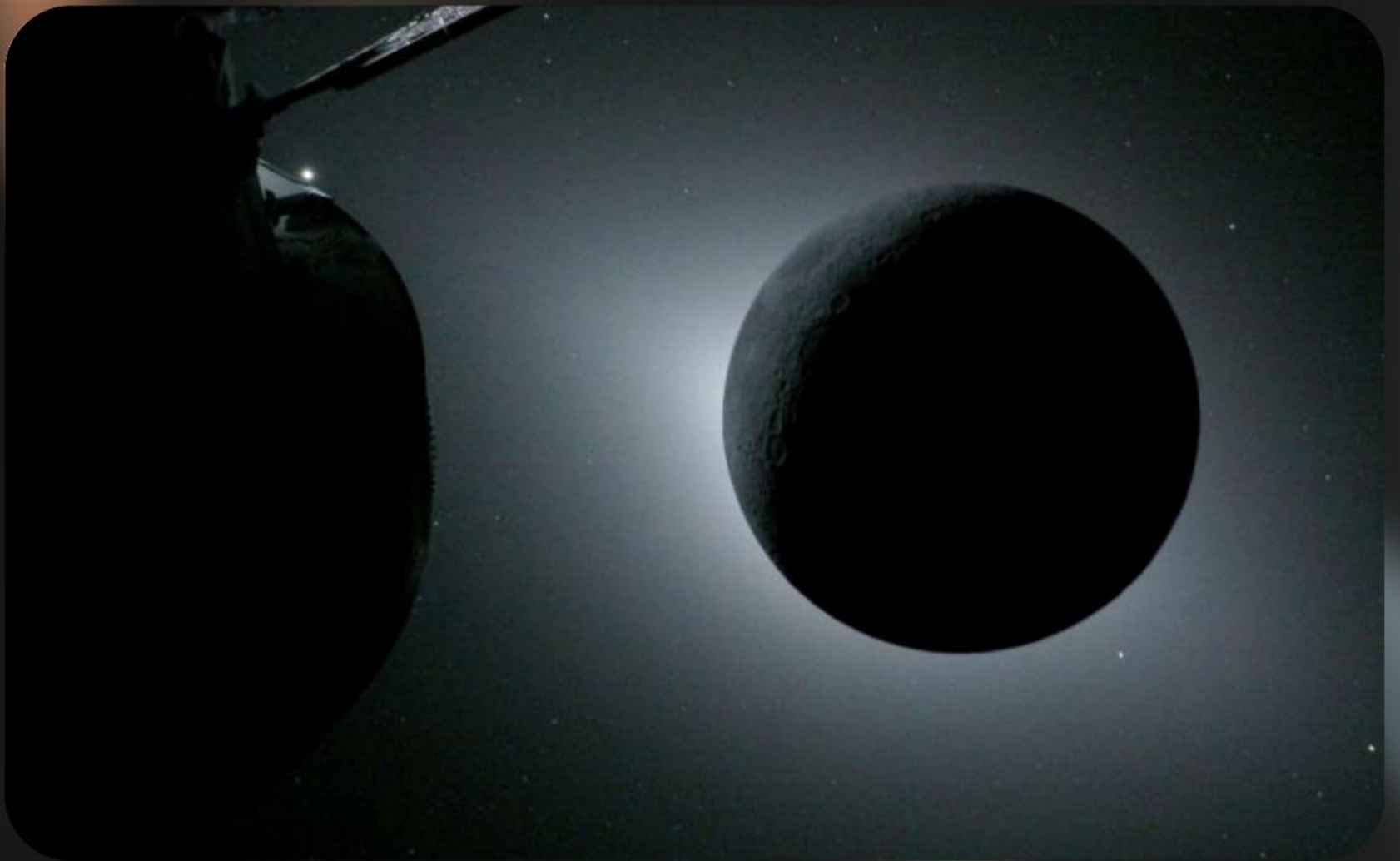
## Crew and International Participation

The four crew members include Victor Glover, the first person of colour; Christina Koch, the first woman; Canadian Space Agency astronaut Jeremy Hansen, the first non-U.S. citizen; and Commander Reid Wiseman, the oldest person to travel beyond low Earth orbit and around the Moon.





Earthset captured through the Orion spacecraft window at 6:41 p.m. EDT, April 6, 2026, during the Artemis II crew's flyby of the Moon



The Moon, seen here backlit by the Sun during a solar eclipse on April 6, 2026, is photographed by one of the cameras on the Orion spacecraft's solar array wings.



## Profile of the Flight

Artemis II was a crewed flight test in which four astronauts evaluated the performance of the Space Launch System (SLS) rocket, as well as the Orion spacecraft and its European Service Module (ESM), in deep space.

## Challenges of travelling beyond LEO

Artemis II travels beyond Low Earth Orbit (LEO), which normally protects astronauts from space radiation. Astronauts on Artemis II don't have the option of rapid return options unlike International Space Station missions, which don't leave LEO.

Crew members constantly monitor their own cardiovascular health, sleep patterns, and stress functions while relying on autonomous medical kits and procedures.

## Why It Matters?

Artemis II helps establish the presence of humans on or near the Moon, before moving towards more difficult space exploration such as Mars.

• URVI SINGH | 25BCE2014



# PUZZLE

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| E | T | E | V | E | N | T | H | O | R | I | Z | O | N |
| O | B | H | G | V | N | N | D | V | S | E | F | S | E |
| F | Z | E | S | A | C | R | Y | X | H | X | C | Q | S |
| K | R | D | C | O | N | X | N | K | L | H | T | E | U |
| R | K | A | V | N | B | Y | B | I | W | I | V | A | E |
| A | X | K | T | F | A | A | M | A | O | R | W | R | G |
| S | E | D | K | E | J | R | R | E | X | W | F | T | L |
| A | Y | S | G | L | N | Z | E | Y | D | M | I | E | E |
| U | L | X | V | U | S | G | S | V | O | E | B | M | T |
| Q | C | W | W | C | R | Y | A | S | E | N | S | I | E |
| Q | L | I | H | M | L | V | E | M | R | S | I | S | B |
| Q | W | I | G | N | C | N | Q | X | R | L | R | C | X |
| Q | L | B | A | S | T | R | O | S | A | T | B | E | C |
| D | K | D | J | E | Z | H | Y | G | Y | Z | Y | S | P |

Words to look for:

- |                  |                 |
|------------------|-----------------|
| 1. SCHWARZSCHILD | 2. PERSEVERANCE |
| 3. EVENTHORIZON  | 4. BETELGEUSE   |
| 5. GANYMEDE      | 6. BARYONIC     |
| 7. MAGNETAR      | 8. ASTROSAT     |
| 9. ARTEMIS       | 10. SZYZYGY     |
| 11. QUASAR       |                 |



# Stellar Definitions:

1. **Schwarzschild:** Associated with the Schwarzschild solution in general relativity, it is crucial for understanding non-rotating black holes and the Schwarzschild radius, beyond which escape becomes impossible.
2. **Perseverance:** NASA's Mars rover is designed to study the planet's surface, geology, and past environment. It also collects rock samples that can help investigate whether Mars once supported microbial life.
3. **Event Horizon:** The boundary surrounding a black hole where the escape velocity exceeds the speed of light. Once something crosses it, it cannot return or communicate with the outside universe.
4. **Betelgeuse:** A massive red supergiant star in the constellation Orion. It is particularly important for studying the final stages of massive stars and understanding their eventual demise.
5. **Ganymede:** Jupiter's largest moon and the largest natural satellite in the Solar System. It is particularly interesting due to its subsurface ocean and magnetic field.
6. **Baryonic:** Describes ordinary matter primarily composed of baryons such as protons and neutrons. Studying baryonic matter helps us understand stars, galaxies, and the visible matter that constitutes much of the universe.
7. **Magnetar:** A highly magnetised type of neutron star with an extraordinarily strong magnetic field. Magnetars are important for studying extreme matter, magnetic fields, and high-energy cosmic phenomena.
8. **AstroSat:** India's first dedicated multi-wavelength space observatory. It observes celestial objects across various wavelengths, aiding scientists in studying phenomena such as black holes, neutron stars, and distant galaxies.
9. **Artemis:** NASA's programme for lunar exploration and human missions to the Moon. It is significant as a step towards establishing sustained lunar exploration and preparing for future missions to Mars.
10. **Syzygy:** A configuration in which three celestial bodies are approximately aligned. The Sun, Earth, Moon arrangement during a syzygy is particularly important for understanding solar and lunar eclipses.
11. **Quasar:** An extremely bright and distant active galactic nucleus, powered by material falling toward a supermassive black hole. Quasars allow scientists to study black holes, galaxies and conditions in the early universe.



# SOLUTIONS TO PUZZLE

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| E | T | E | V | E | N | T | H | O | R | I | Z | O | N |
| O | B | H | G | V | N | N | D | V | S | E | F | S | E |
| F | Z | E | S | A | C | R | Y | X | H | X | C | Q | S |
| K | R | D | C | O | N | X | N | K | L | H | T | E | U |
| R | K | A | V | N | B | Y | B | I | W | I | V | A | E |
| A | X | K | T | F | A | A | M | A | O | R | W | R | G |
| S | E | D | K | E | J | R | R | E | X | W | F | T | L |
| A | Y | S | G | L | N | Z | E | Y | D | M | I | E | E |
| U | L | X | V | U | S | G | S | V | O | E | B | M | T |
| Q | C | W | W | C | R | Y | A | S | E | N | S | I | E |
| Q | L | I | H | M | L | V | E | M | R | S | I | S | B |
| Q | W | I | G | N | C | N | Q | X | R | L | R | C | X |
| Q | L | B | A | S | T | R | O | S | A | T | B | E | C |
| D | K | D | J | E | Z | H | Y | G | Y | Z | Y | S | P |



# CONTRIBUTERS

## DESIGNERS

Rasheswari Padhi  
Aadit A Athawase  
Darsh L  
Swastik Naskar  
Abhinav Kumar

## EDITORS

Nakshatra Gundeli  
Ishan Saxena  
Neha Gireesh  
M Harinivethitha  
Urvi Singh  
N. Sanjanaa  
Evelyn Renee R  
Harinipriya K  
Anirudh Balaji  
Sahila Yazar